

Education

- 2018–2022 **The Chinese University of Hong Kong**,
Ph.D. in Computer Science and Engineering,
Supervisor: *James Cheng*.
- 2013–2017 **South China University of Technology**,
B.Eng in Computer Science and Technology.

Research Interests

Distributed computation, machine learning framework, cluster scheduling.

Work Experiences

- Oct 2021 - **Amazon Web Service, AI Lab**.
Ongoing
 - Applied Scientist Intern in DGL Team
 - Distributed sampling on GPUs for GNN training
- Nov 2020 - **Alibaba Group, Apsara Platform**.
Sept 2021
 - Research Intern in Fuxi Team
 - Resource management for clusters with thousands of GPUs
- July 2019 - **HUAWEI, 2012 Lab**.
Sept 2020
 - Research Intern in Mindspore Team
 - Auto-parallelism DNN training systems for large model training
 - Efficient distributed GNN training systems
- July 2018 - **Alibaba Group, Apsara Platform**.
Aug 2018
 - Research Intern in MaxCompute Team
 - Smart index selection for large tables in data warehouses
- July 2017 - **The Chinese University of Hong Kong**.
July 2018
 - Research Assistant supervised by Prof. James Cheng
 - Distributed Systems for Large Scale Data Analytics

Awards

- 2018 - 2022 **Postgraduate Studentship, CUHK**.
- 2014 - 2016 **Undergraduate National Scholarship, SCUT**.
- 2014 - 2016 **Merit Student, SCUT**.
- 2017 **ACM-ICPC World Finalist**.
- 2016 **ACM-ICPC Regional Contest Qingdao Site 2016, Gold Medal**.
- 2016 **ACM-ICPC Regional Contest Shenyang Site 2016, Gold Medal**.

Skills

- Programming **C/C++, Python, CUDA**.
- Systems **MPI, Tensorflow, Pytorch, NCCL**.

Publications

- [1] **PPS: Fair and Efficient Scheduling for Multi-Tenant GPU Clusters**,
Z. Cai, K. Ma, X. Yan, Y. Zhang, Z. Liu, Y. Feng, C. Li, J. Cheng,
NSDI 2022, Under Review.

- [2] **DGCL: An Efficient Communication Library for Distributed GNN Training**,
Z. Cai, X. Yan, Y. Wu, K. Ma, J. Cheng, F. Yu,
EUROSYS 2021.
- [3] **Seastar: Vertex-Centric Programming for Graph Neural Networks**,
Y. Wu, K. Ma, Z. Cai, T. Jin, B. Li, C. Zheng, J. Cheng, F. Yu,
EUROSYS 2021.
- [4] **Elastic Deep Learning in Multi-Tenant GPU Clusters**,
Y. Wu, K. Ma, X. Yan, Z. Liu, Z. Cai, Y. Huang, J. Cheng, H. Yuan, F. Yu,
TPDS 2021.
- [5] **TensorOpt: Exploring the Tradeoffs in Distributed DNN Training with Auto-Parallelism**,
Z. Cai, K. Ma, X. Yan, Y. Wu, Y. Huang, J. Cheng, T. Su, F. Yu,
TPDS 2021, Under Review.
- [6] **Improving Resource Utilization by Timely Fine-Grained Scheduling**,
T. Jin, Z. Cai, B. Li, C. Zheng, G. Jiang, J. Cheng,
EUROSYS 2020.
- [7] **FlexPS: Flexible Parallelism Control in Parameter Server Architecture**,
Y. Huang, T. Jin, Y. Wu, Z. Cai, X. Yan, Y. Guo, F. Yang, J. Li, and J. Cheng,
VLDB 2018.
- [8] **Scalable De Novo Genome Assembly Using Pregel**,
D. Yan, H. Chen, Z. Cai, J. Cheng, and B. Shao,
ICDE 2018.

Projects

Systems for large-scale machine learning

TensorOpt: A Distributed DNN Training System on GPUs.

- Supports large model training on limited GPU memory
- Optimally decides the parallelization strategies of each operator in a DNN
- User-friendly APIs

EDL: An Elastic Deep Learning System on GPUs.

- Simple APIs supporting elastic deep learning on GPUs
- Stop-free scaling and dynamic data pipeline

FlexPS: A Parameter Server with Flexible Parallelism Control.

- A novel multi-stage abstraction to support flexible parallelism control
- Optimizations for efficient parallelism adjustment

Systems for graph neural networks

DGCL: A Distributed Graph Communication Library for GNN systems.

- A general library to scale single-GPU GNN systems to the multi-GPU setting
- Efficient communication kernels optimizing for load balancing and bandwidth contention on NVLinks, PCIe and InfiniBand

Seastar: A Vertex-centric GNN System.

- A vertex-centric programming model for GNNs
- Kernel optimizations such as operator fusion and vertex parallelism to reduce GPU memory consumption and training time

Large-scale cluster scheduling

PPS: Fair And Efficient Scheduling for Multi-Tenant GPU Clusters.

- Probabilistic prediction based scheduling for clusters with thousands of GPUs
- Black-box and non-preemptive scheduling for GPU jobs
- Achieves both efficiency and fairness at the same time

Ursa: A Framework for both Resource Scheduling and Job Execution.

- Captures dynamic resource needs at runtime and enables fine-grained, timely scheduling
- Achieves high resource utilization, which is translated into significantly improved makespan and average JCT